Bias and Fairness

Roadmap

Introduction

Types and Sources of Bias in AI Systems

Algorithmic Fairness and Non-discrimination

Techniques for Mitigating Bias in AI Models

Introduction

Key Concepts and Definitions

- AI bias: refers to systematic errors or unfairness in AI systems that can lead to discriminatory outcomes or decisions
- Fairness in AI: ensures that AI systems treat all individuals or groups equitably without discrimination based on protected attributes
- Protected attributes: refer to characteristics that should not influence AI decision-making to prevent discrimination (e.g., ethnicity, gender, age, disability status)
- Algorithmic fairness: aims to develop AI systems that make decisions without discriminating against certain groups or individuals

Impact on Different Communities

- AI bias can perpetuate and exacerbate existing societal inequalities and discrimination faced by marginalized communities
- Biased AI systems in hiring and lending can limit economic opportunities and upward mobility for underrepresented groups
- Biased facial recognition systems can lead to false arrests and wrongful convictions, disproportionately affecting communities of color

Impact on Different Communities (2)

- AI bias in healthcare can result in disparities in medical treatment, misdiagnosis, and poorer health outcomes for certain populations
- AI bias in education can reinforce achievement gaps and limit access to resources and opportunities for disadvantaged students

Impact on Different Communities (3)

- AI bias in content recommendation systems can limit exposure to diverse perspectives and reinforce echo chambers or extremist views
- AI bias in criminal justice can lead to harsher sentencing and perpetuate cycles of incarceration for already overrepresented groups

Types and Sources of Bias in AI Systems

Common Types of Bias

- Algorithmic bias: causes systematic errors in AI systems leading to unfair outcomes
- Selection bias: occurs when training data misrepresents the target population
- Sampling bias: results from over- or under-representation of certain groups in data
- While all can lead to flawed outcomes, they stem from different processes and have different implications for data analysis and decision-making

Sources of AI Bias

- Data collection methods introduce bias to give imbalanced training data
- Data labeling processes introduce human biases
- Feature selection emphasizes or de-emphasizes certain attributes
- Proxy variables correlate with protected attributes, introducing unintended bias
- Developer biases encoded during system development
- Feedback loops in deployed systems amplify existing biases over time

Impact of Biased AI

- Biased AI in hiring perpetuates workplace inequalities
- Credit scoring and loan approval systems limit financial opportunities
- Facial recognition systems lead to false identifications
- Automated decision-making in criminal justice perpetuates systemic racism

Impact of Biased AI (2)

- Biased AI in healthcare results in misdiagnoses and inadequate treatment
- Content recommendation systems create echo chambers and filter bubbles
- Large language models generate and amplify stereotypes

Real-World AI Bias Examples

- COMPAS recidivism prediction tool exhibited racial bias in risk assessments
- Facial recognition systems used by law enforcement show lower accuracy for minorities
- Amazon's experimental AI recruiting tool showed bias against women candidates
- Apple Card credit limit controversy revealed gender bias in financial algorithms

Real-World AI Bias Examples (2)

- Google Photos image recognition system mislabeled Black people as “gorillas”
- Healthcare AI systems perform less accurately on darker skin tones

Algorithmic Fairness and Non-discrimination

What is Algorithmic Fairness?

- Algorithmic fairness ensures absence of bias or discrimination in automated decision-making systems, promoting equitable outcomes across demographic groups
- Maintains public trust, legal compliance, and ethical integrity

Different Notions of Algorithmic Fairness

- Individual fairness: Similar individuals should receive similar outcomes, regardless of group
- Group fairness: An algorithm's predictions should be consistent or balanced across different demographic groups (e.g., racial groups, genders)
- Counterfactual fairness: A decision is fair if it would remain the same if an individual's sensitive attribute (e.g., race) were different, holding all other factors constant

Fairness Metrics

Fairness metrics measure different aspects of model predictions across protected groups:

- Demographic parity
- Equal opportunity
- Equalized odds

Challenges of Algorithmic Fairness

- Inherent trade-offs between different fairness definitions
- Complexity of real-world scenarios
- Potential for unintended consequences
- Multidisciplinary approach needed involving:
  - Computer scientists
  - Ethicists
  - Legal experts
  - Domain specialists

Strategies for Ensuring Non-discrimination

- “Fairness through unawareness” by removing protected attributes from data often insufficient due to correlated features
- Algorithmic transparency and explainability crucial for identifying and addressing discriminatory patterns
- Proactive measures for non-discrimination
  - Diverse and representative training data
  - Regular bias audits
  - Implementation of bias mitigation techniques throughout AI lifecycle

Trade-offs in Model Design

- "Impossibility theorem" demonstrates that multiple fairness criteria cannot be satisfied simultaneously
- Accuracy-fairness trade-offs occur when improving fairness metrics may decrease overall model accuracy
- Pareto optimality analyzes and visualizes trade-offs between multiple objectives (fairness, accuracy)

Trade-offs in Model Design (2)

- Techniques for managing trade-offs:
  - Multi-objective optimization
  - Constrained optimization
  - Adversarial debiasing
- Each technique has unique strengths and limitations

Navigating Trade-offs

- Choice of fairness metric impacts other performance measures
- Depends on specific context, stakeholder priorities, and ethical considerations of AI application
- Requires consideration of long-term societal impacts, not just immediate performance metrics
- Ensures sustainable and ethical AI deployments
- Process involves
  - Iterative refinement
  - Stakeholder engagement
  - Ongoing monitoring

Techniques for Mitigating Bias in AI Models

Unfairness Reduction Approaches

Data preprocessing

Smart model design

Postprocessing

Data Resampling and Augmentation

- Resampling methods balance representation in imbalanced datasets
  - Oversampling increases instances of minority classes
  - Undersampling reduces instances of majority classes
- Data augmentation increases diversity of training data
  - Generates synthetic examples for underrepresented groups
  - Applies transformations (e.g., rotation, translation) to existing data points

Feature Engineering and Data Cleaning

- Feature selection identifies and removes bias-introducing features
- Feature engineering modifies existing features to reduce bias
- Data cleaning ensures consistent representation across groups
- Normalization techniques reduce unintended biases in numerical features

Bias Detection and Mitigation Algorithms

- Bias detection algorithms identify potential sources of bias in training data
- Fairness constraints incorporated during data preprocessing
- Causal inference techniques used to understand and mitigate bias

Regularization and Ensemble Methods

- Regularization techniques prevent overfitting to biased patterns
- Ensemble methods combine multiple diverse models to reduce bias
- Dropout layers in neural networks promote robustness to bias

Adversarial Debiasing and Multi-Task Learning

- Adversarial debiasing explicitly optimizes for fairness during training
- Multi-task learning jointly optimizes for performance and fairness
- Fair representation learning creates unbiased intermediate representations

Transfer Learning and Fine-Tuning Strategies

- Transfer learning leverages pre-trained models while adapting for fairness
- Domain adaptation techniques adjust for differences in data distributions
- Continual learning approaches maintain fairness as new data arrives

Threshold Adjustment and Calibration

- Threshold adjustment balances error rates across groups
- Calibration methods align model confidence with true probabilities
- Reject option classification handles potentially biased decisions

Reweighting and Equalized Odds

- Reweighting algorithms adjust predictions based on group membership
- Equalized odds postprocessing achieves similar error rates across groups
- Fairness constraints incorporated into decision-making process

Counterfactual Fairness and Causal Approaches

- Counterfactual fairness techniques adjust outputs based on causal relationships
- Causal inference methods applied to understand and mitigate unfairness
- Fairness through unawareness removes protected attributes from decision process

Case Studies

Case Studies (Stahl et al. 2023)

- Chapter 2, Case 1: Gender bias in recruitment tools
- Chapter 2, Case 2: Discriminatory use of AI in law enforcement and predictive policing
- Chapter 2, Case 3: Discrimination on the basis of skin color